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Department

Of

Electronics and Communication Engineering

Minor Project Report on

Radiation Hardened Techniques to mitigate SEUs

By: Team 09

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K.L.E SOCIETY'S

KLE Technological University,
HUBBALLI-580031 2025-26



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that project entitled **"Radiation Hardened Techniques to mitigate SEUs"** is a Bonafide work carried out by the student team of **Jason Anthoni (01FE23BEC076), Vaishnavi Srekar(01FE23BEC318), Rahul Kavaloor(01FE23BEC090), Vidyadhar Shenvi(01FE23BEC066)**. The project report has been approved as it satisfies the requirements with respect to the Mini project work prescribed by the university curriculum for BE (V semester) in Department of Electronics and Communication Engineering of KLE Technological University for the academic year 2025-26.

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Name of Examiners	**Signature with date**
1.	
2.	

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-The project team

ABSTRACT

As semiconductor technology continues to scale into deep submicron dimensions, Static Random Access Memory (SRAM) cells have become increasingly susceptible to radiation-induced soft errors. Among these, the Single Event Upset (SEU) is one of the most significant reliability challenges, particularly in aerospace, satellite, automotive, and safety-critical electronic systems. An SEU occurs when an energetic particle such as a neutron, alpha particle, or heavy ion strikes a sensitive node of a memory cell, depositing sufficient charge to alter the stored logic state without causing permanent physical damage. Consequently, the development of radiation-hardened memory architectures has become essential for ensuring reliable operation in harsh environments.

The primary objective of this project is to investigate the vulnerability of a conventional 6-Transistor (6T) SRAM cell to Single Event Upsets and to evaluate the effectiveness of the Dual Interlocked Storage Cell (DICE) architecture in mitigating these radiation-induced errors. Initially, a conventional 6T SRAM cell was designed and simulated using the Cadence Virtuoso design environment. The functional correctness of the SRAM was verified by performing read, write, and hold operations. A precharge circuit was also designed and integrated to facilitate proper bit-line operation during memory access.

To emulate the effects of radiation, a transient current pulse was injected at the sensitive storage node of the conventional SRAM cell. By gradually varying the magnitude of the injected current, the critical charge threshold required to induce a Single Event Upset was determined. The simulation results demonstrated that once the injected charge exceeded the critical value, the stored data in the conventional 6T SRAM cell flipped, confirming its susceptibility to radiation-induced soft errors.

Subsequently, a DICE SRAM cell was designed and simulated under identical operating conditions. The same transient current pulse was injected into one of its storage nodes to evaluate its robustness against SEUs. Unlike the conventional SRAM, the DICE architecture successfully restored the original stored data after the transient disturbance, demonstrating its inherent fault-tolerant capability. The interlocked storage mechanism prevented permanent data corruption by allowing the unaffected nodes to reinforce the correct logic state.

The results obtained from this work clearly demonstrate that while the conventional 6T SRAM is vulnerable to Single Event Upsets beyond its critical charge threshold, the DICE SRAM architecture provides significantly improved resilience against radiation-induced disturbances. This study highlights the importance of radiation-hardened-by-design (RHBD) techniques in enhancing memory reliability and provides a comparative analysis of conventional and hardened SRAM architectures for future high-reliability integrated circuit applications.

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CHAPTER 1

INTRODUCTION

1.1 Background

The continuous advancement of semiconductor technology has enabled the fabrication of integrated circuits with billions of transistors on a single chip. CMOS (Complementary Metal-Oxide-Semiconductor) technology has become the dominant fabrication process due to its low power consumption, high switching speed, and excellent scalability. As technology nodes continue to shrink, transistors become smaller, faster, and more energy-efficient, allowing modern electronic systems to achieve remarkable computational performance while occupying minimal silicon area.

Static Random Access Memory (SRAM) is one of the most widely used semiconductor memories in modern digital systems. It serves as cache memory in microprocessors, embedded memory in System-on-Chip (SoC) devices, and temporary storage in numerous digital applications. SRAM is preferred because of its high operating speed, low access latency, and compatibility with CMOS fabrication processes. The conventional 6-Transistor (6T) SRAM cell has remained the standard memory architecture due to its simplicity and efficiency.

However, as CMOS technology scales into deep submicron and nanometer dimensions, the amount of charge stored at the internal nodes of an SRAM cell decreases significantly. This reduction makes memory cells increasingly vulnerable to external disturbances such as radiation-induced charge collection. Even a relatively small amount of deposited charge can alter the stored logic value, thereby reducing the overall reliability of the memory system.

The demand for highly reliable memories has grown rapidly due to their extensive use in critical applications. In aerospace systems, satellites and spacecraft are continuously exposed to cosmic radiation outside the Earth's atmosphere. Automotive electronics, particularly in autonomous driving and advanced driver assistance systems (ADAS), require dependable memories to ensure passenger safety under varying environmental conditions. Medical devices, including implantable electronics and diagnostic equipment, must maintain accurate operation to prevent life-threatening failures. Similar reliability requirements exist in defence, industrial automation, and nuclear power monitoring systems.

These challenges have motivated researchers to develop radiation-hardened memory architectures capable of maintaining data integrity even in harsh operating environments. Radiation-Hardened-By-Design (RHBD) techniques provide an effective solution by modifying circuit architectures rather than relying solely on specialized fabrication technologies.

1.2 Motivation

The increasing integration density of modern integrated circuits has significantly improved performance and reduced manufacturing costs. However, this scaling has also increased the susceptibility of semiconductor devices to radiation-induced soft errors. As supply voltages decrease and transistor dimensions shrink, the critical charge required to change the logic state of a memory cell becomes smaller. Consequently, even naturally occurring radiation can disturb the stored information.

Energetic particles originating from cosmic rays, alpha particle emissions from packaging materials, and secondary neutrons generated in the Earth's atmosphere are capable of penetrating semiconductor devices. When these particles strike sensitive regions of a transistor, they generate electron-hole pairs that result in transient current pulses. If the collected charge exceeds the critical charge of the memory cell, the stored logic value may invert, producing a Single Event Upset (SEU).

Although SEUs do not permanently damage the physical structure of the device, they may lead to corrupted data, incorrect computations, system crashes, or unexpected behavior in safety-critical applications. The consequences are particularly severe in satellites, spacecraft, military electronics, aviation systems, medical equipment, and autonomous vehicles, where system reliability is of paramount importance.

Traditional approaches such as Error Correcting Codes (ECC) and Triple Modular Redundancy (TMR) improve reliability but introduce additional hardware complexity, increased area, and higher power consumption. Radiation-Hardened-By-Design (RHBD) techniques provide an attractive alternative by improving the intrinsic robustness of memory cells. Among these techniques, the Dual Interlocked Storage Cell (DICE) architecture has emerged as one of the most effective solutions for mitigating Single Event Upsets while maintaining reliable operation.

The motivation for this project is to study the susceptibility of a conventional 6T SRAM cell to radiation-induced disturbances and to evaluate the ability of the DICE SRAM architecture to recover from transient charge injections.

1.3 Problem Statement

Conventional 6T SRAM cells are designed primarily to achieve high speed, low power consumption, and compact area. However, they were not originally intended to withstand radiation-induced transient disturbances. As semiconductor technologies continue to scale, the stored charge within the internal storage nodes decreases, making these nodes increasingly sensitive to externally induced charge injection.

When a high-energy particle strikes a sensitive node within a conventional SRAM cell, it deposits charge that may temporarily alter the node voltage. If the deposited charge exceeds the critical charge of the memory cell, the positive feedback mechanism of the cross-coupled inverters forces the cell into the opposite stable state, resulting in a Single Event Upset. This phenomenon leads to data corruption without causing permanent physical damage to the device.

For applications operating in radiation-rich environments, such as satellites, aerospace electronics, and safety-critical control systems, conventional SRAM cells may not provide sufficient reliability. Therefore, there is a need for radiation-hardened memory architectures capable of resisting or automatically recovering from radiation-induced disturbances without external intervention.

This project addresses this challenge by evaluating the radiation tolerance of a conventional 6T SRAM cell and comparing its performance with a DICE SRAM architecture through transient current injection simulations.

1.4 Objectives

The primary objective of this project is to investigate radiation-induced Single Event Upsets in SRAM cells and evaluate a radiation-hardened memory architecture capable of mitigating these effects.

The specific objectives are as follows:

- To design and implement a conventional 6T SRAM cell using CMOS technology.
- To verify the functionality of the SRAM cell through read, write, and hold operations.
- To design a precharge circuit for proper bit-line operation during memory access.
- To simulate radiation-induced charge deposition by injecting transient current pulses into the SRAM cell.
- To determine the critical charge threshold at which a Single Event Upset occurs in the conventional SRAM.
- To design and implement a Dual Interlocked Storage Cell (DICE) SRAM architecture.
- To perform transient current injection on the DICE SRAM cell under identical operating conditions.
- To compare the performance of the conventional 6T SRAM and DICE SRAM in terms of radiation tolerance and data stability.
- To demonstrate the effectiveness of Radiation-Hardened-By-Design techniques in improving memory reliability.

1.5 Scope

This project focuses on the design and simulation-based analysis of radiation-induced Single Event Upsets in SRAM cells. The work is limited to schematic-level implementation and transient simulations using the Cadence Virtuoso design environment. A conventional 6T SRAM cell is first designed and validated through read, write, and hold

operations, followed by the implementation of a precharge circuit to facilitate memory access.

The effects of radiation are emulated by injecting transient current pulses into the storage nodes of the SRAM cell to determine the critical charge required for data corruption. A DICE SRAM architecture is then designed and subjected to identical fault injection conditions to evaluate its capability to recover from transient disturbances.

The scope of this project is restricted to simulation-based verification and does not include physical fabrication, post-layout analysis, or experimental radiation testing using particle accelerators. Nevertheless, the results provide valuable insight into the effectiveness of radiation-hardened memory architectures and establish a foundation for future work involving layout optimization, process variation analysis, FinFET technologies, and hardware implementation.

The techniques investigated in this project have potential applications in aerospace electronics, satellite communication systems, automotive safety controllers, medical electronics, industrial automation, military systems, and other environments where memory reliability is a critical requirement.

CHAPTER 2

LITERATURE REVIEW

2.1 Static Random Access Memory (SRAM)

Static Random Access Memory (SRAM) is a volatile semiconductor memory widely used in digital systems due to its high operating speed and low access latency. Unlike Dynamic Random Access Memory (DRAM), SRAM stores data using bistable latching circuitry and does not require periodic refreshing to retain information as long as power is supplied.

The earliest SRAM cells were developed during the late 1960s using bipolar transistor technology. With the rapid advancement of CMOS technology, SRAM became the preferred memory for on-chip cache memories because of its lower power consumption, higher integration density, and compatibility with modern VLSI fabrication processes. Today, SRAM is extensively employed as Level-1 (L1), Level-2 (L2), and Level-3 (L3) cache memory in microprocessors, embedded memory in microcontrollers, digital signal processors, Field Programmable Gate Arrays (FPGAs), and System-on-Chip (SoC) devices.

The most common SRAM architecture is the 6-Transistor (6T) SRAM cell, consisting of two cross-coupled CMOS inverters and two access transistors. The cross-coupled inverters form a bistable latch capable of storing one bit of information, while the access transistors connect the storage nodes to the complementary bit lines during read and write operations.

As CMOS technology scales below nanometer dimensions, transistor sizes and supply voltages continue to decrease. Consequently, the amount of charge stored at the internal storage nodes also decreases, making SRAM cells increasingly vulnerable to process variations, noise, and radiation-induced soft errors.

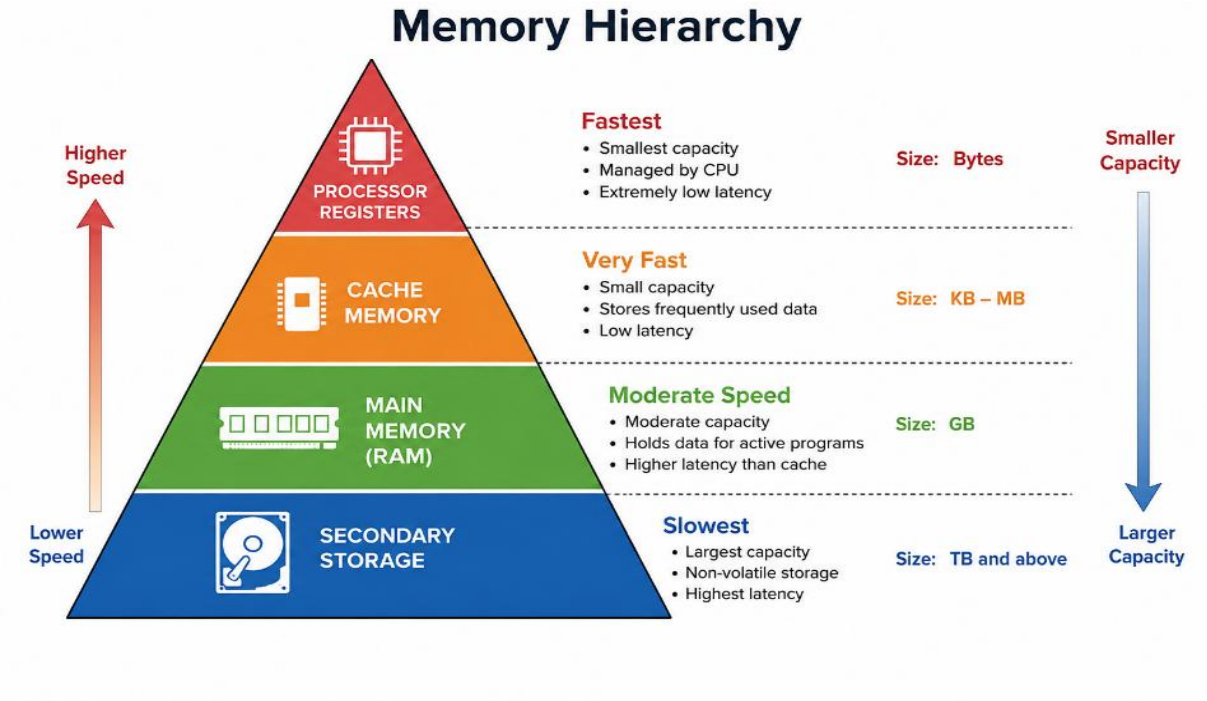


Figure 2.1: Memory Hierarchy (Processor Registers → Cache → Main Memory → Secondary Storage)

2.2 Radiation Effects in Semiconductor Devices

Semiconductor devices operating in terrestrial environments, aircraft, satellites, and deep-space missions are continuously exposed to ionizing radiation. High-energy particles interact with semiconductor materials, generating electron-hole pairs that may temporarily or permanently alter device characteristics.

Radiation effects are generally classified into two categories:

2.2.1 Total Ionizing Dose (TID)

Total Ionizing Dose refers to the cumulative damage caused by prolonged exposure to ionizing radiation. As radiation accumulates over time, trapped charges are formed inside the gate oxide and at the oxide-semiconductor interface. These trapped charges alter transistor threshold voltages, increase leakage current, reduce carrier mobility, and eventually degrade circuit performance.

TID effects are gradual and primarily affect long-term device reliability rather than immediate functionality.

2.2.2 Single Event Effects (SEE)

Single Event Effects occur when a single energetic particle strikes a semiconductor device. Unlike TID, these effects are instantaneous and may or may not cause permanent damage.

Depending on the interaction, Single Event Effects are categorized into several types.

2.2.3 Single Event Upset (SEU)

A Single Event Upset is a transient change in the stored logic state of a memory element caused by the collection of radiation-induced charge. The upset does not damage the physical structure of the device but results in incorrect stored data.

SEUs are commonly referred to as soft errors because they can usually be corrected by rewriting the correct data.

2.2.4 Single Event Transient (SET)

A Single Event Transient is a temporary voltage or current pulse generated in combinational logic when an energetic particle strikes a transistor. If this transient propagates through the logic and reaches a storage element during a clock transition, it may eventually become an SEU.

2.2.5 Single Event Latch-up (SEL)

Single Event Latch-up is a potentially destructive radiation effect in CMOS circuits. It occurs when parasitic PNPN structures inside the CMOS substrate are triggered into conduction, creating a low-resistance path between the power supply and ground.

SEL results in excessive current flow and may permanently damage the integrated circuit if not interrupted immediately.

2.3 Single Event Upset (SEU)

Among all radiation-induced soft errors, the Single Event Upset represents one of the greatest reliability concerns for modern SRAM-based systems.

When a high-energy particle such as an alpha particle, neutron, or heavy ion passes through silicon, it loses energy through ionization. This process generates thousands of electron-hole pairs along its trajectory. Under the influence of the electric field present at the reverse-biased drain junctions of MOS transistors, these charge carriers are rapidly collected, producing a transient current pulse.

If the collected charge exceeds the critical charge (Qcrit) required to change the stored state of the memory cell, the cross-coupled inverters reinforce the new logic level, permanently flipping the stored bit until corrected.

The susceptibility of an SRAM cell to SEU depends upon several factors, including:

- Technology node
- Supply voltage
- Stored charge
- Junction capacitance
- Particle energy
- Pulse duration
- Device geometry

As CMOS technology scales, the critical charge decreases significantly, making advanced SRAM technologies increasingly susceptible to radiation-induced soft errors.

2.4 Radiation Hardened Design Techniques

Several techniques have been developed to improve the radiation tolerance of semiconductor memories. These methods differ in complexity, hardware overhead, performance, and effectiveness.

2.4.1 Triple Modular Redundancy (TMR)

Triple Modular Redundancy replicates the same circuit three times and determines the correct output through majority voting. If one module experiences an SEU, the remaining two modules maintain the correct output.

Although TMR provides excellent fault tolerance, it introduces significant area, power, and hardware overhead.

2.4.2 Error Correcting Codes (ECC)

Error Correcting Codes add redundant parity bits to stored data, allowing detection and correction of bit errors during memory access.

ECC is widely used in server processors, aerospace systems, and mission-critical computing platforms. However, it increases memory overhead and access latency.

2.4.3 Hardened Storage Elements

Radiation-hardened latches modify conventional latch architectures to increase their immunity against transient disturbances.

Examples include Dual Interlocked Storage Cells (DICE), Quatro latches, and other hardened storage structures designed specifically for radiation environments.

2.4.4 Radiation-Hardened-by-Design (RHBD)

Radiation-Hardened-by-Design improves circuit robustness through architectural and layout modifications without requiring specialized fabrication technologies.

RHBD techniques include:

- Guard rings
- Enclosed layout transistors
- Node separation
- Redundant storage
- Capacitive hardening
- DICE memory cells

RHBD is widely adopted because it provides enhanced radiation tolerance while remaining compatible with standard CMOS manufacturing processes.

2.4.5 Dual Interlocked Storage Cell (DICE)

The DICE architecture is one of the most successful Radiation-Hardened-by-Design techniques for SRAM applications.

Unlike the conventional 6T SRAM cell, which stores information using two storage nodes, the DICE cell stores information using four mutually interlocked storage nodes. This redundancy ensures that if a transient disturbance flips one node, the remaining nodes automatically restore the original data through positive feedback.

As a result, a single particle strike is generally insufficient to corrupt the stored information permanently, significantly improving memory reliability in radiation-prone environments.

2.5 Previous Research

Extensive research has been conducted to improve the radiation tolerance of SRAM cells through both circuit-level and architectural techniques.

Recent work by **Oh and Jo (2024)** proposed a radiation-hardened 16T SRAM cell that demonstrated improved read stability and enhanced resistance to Single Event Upsets

for space applications. Although the design significantly increased reliability, the additional transistors resulted in increased silicon area.

Zhang et al. (2023) investigated radiation-hardened SRAM-FPGA systems using multi-level fault mitigation techniques. Their work demonstrated improved system reliability through multiple layers of protection but introduced greater implementation complexity.

Ming et al. (2017) analyzed the Single Event Upset cross-section of DICE SRAM cells and experimentally verified their superior tolerance compared with conventional SRAM architectures.

Gorbunov et al. (2015) proposed a semi-empirical method for estimating the SEU cross-section of DICE SRAM cells. Their methodology reduced computational effort while maintaining good prediction accuracy.

More recently, **Xiong et al. (2025)** compared various Radiation-Hardened-by-Design techniques implemented in a 3 nm Bulk FinFET technology. Their study concluded that architectural hardening methods such as DICE remain highly effective even in advanced technology nodes.

These studies collectively demonstrate that while conventional SRAM cells remain susceptible to radiation-induced soft errors, architectures incorporating redundant storage mechanisms offer significantly improved resilience.

Table 2.1 Previous Research Summary

Authors	Method	Major Result	Limitation
Zhang et al. (2023)	Multi-Level RHBD for SRAM FPGA	Improved reliability	Increased design complexity
Oh & Jo (2024)	Radiation Hardened 16T SRAM	Better read/write stability	Larger area overhead
Ming et al. (2017)	DICE SRAM SEU Analysis	High SEU tolerance	Experimental setup required
Gorbunov et al. (2015)	Semi-empirical DICE Model	Fast SEU estimation	Model approximation
Xiong et al. (2025)	RHBD in 3 nm FinFET	Excellent radiation tolerance	Advanced fabrication required

CHAPTER 3

THEORY

3.1 CMOS Basics

Complementary Metal-Oxide-Semiconductor (CMOS) technology is the foundation of modern Very Large Scale Integration (VLSI) systems due to its low power consumption, high noise immunity, and excellent scalability. CMOS circuits are constructed using two complementary transistor types: the N-channel Metal-Oxide-Semiconductor (NMOS) transistor and the P-channel Metal-Oxide-Semiconductor (PMOS) transistor.

An NMOS transistor conducts when its gate voltage is higher than its threshold voltage. It provides a low-resistance path between the drain and source terminals, making it highly effective in pulling the output node towards ground (logic '0'). Electrons act as the majority charge carriers in NMOS devices, resulting in higher carrier mobility and faster switching speeds.

Conversely, a PMOS transistor conducts when its gate voltage is lower than its threshold voltage. It connects the output node to the supply voltage (VDD), thereby producing a logic '1'. Holes serve as the majority carriers in PMOS devices, which generally have lower mobility than electrons.

The complementary operation of NMOS and PMOS transistors enables CMOS circuits to consume extremely low static power because, ideally, only one transistor conducts during steady-state operation. This characteristic has made CMOS the preferred technology for designing high-performance digital integrated circuits, including SRAM cells.

The fundamental building block of CMOS logic is the CMOS inverter, which consists of one PMOS transistor connected to the power supply and one NMOS transistor connected to ground. The output is taken from the common drain connection of both transistors. When the input is low, the PMOS conducts while the NMOS remains OFF, producing a logic '1'. When the input is high, the NMOS conducts while the PMOS switches OFF, resulting in a logic '0'.

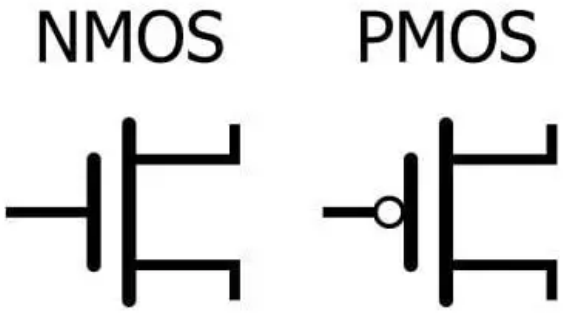


Figure 3.1: NMOS and PMOS transistor symbols and structures.

3.2 SRAM Cell

Static Random Access Memory (SRAM) stores binary information using bistable latching circuits rather than capacitors. The most widely used SRAM architecture is the conventional 6-Transistor (6T) SRAM cell due to its simplicity, speed, and low power consumption.

The 6T SRAM cell consists of:

- Two cross-coupled CMOS inverters.
- Two access transistors.
- Two complementary bit lines (BL and BLB).
- One word line (WL).

The two cross-coupled inverters create positive feedback, producing two stable operating states that represent logic '0' and logic '1'. Once a state is established, the positive feedback continuously reinforces the stored value as long as power is supplied.

The access transistors connect the storage nodes to the bit lines whenever the word line is asserted. During write operations, the bit lines force the desired logic into the storage nodes. During read operations, the stored information slightly modifies the voltage on one of the bit lines, allowing external sense amplifiers to determine the stored data.

The stability of the SRAM cell depends on the transistor sizing ratio between the pull-up, pull-down, and access transistors. Proper sizing ensures reliable read, write, and hold operations while maintaining adequate noise margins.

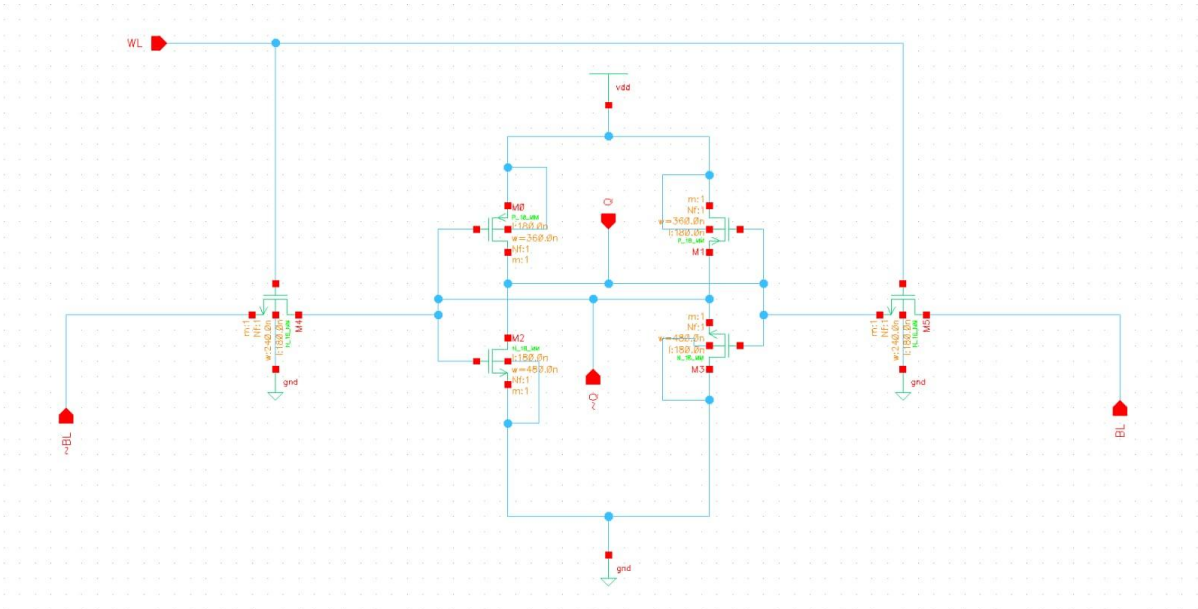


Figure 3.2: 6T SRAM schematic.

3.3 Read Operation

The read operation retrieves the stored information without altering the contents of the memory cell.

Initially, both bit lines are precharged to the supply voltage (VDD) using the precharge circuit. Once the bit lines are fully charged, the word line is activated, turning ON both access transistors.

If the cell stores a logic '1' at node Q and logic '0' at node QB, the node storing logic '0' creates a discharge path through the pull-down NMOS transistor. Consequently, one bit line experiences a slight voltage drop while the complementary bit line remains at VDD.

This small voltage difference is detected by the sense amplifier, which amplifies the difference and determines the stored logic value.

After the read operation is completed, the word line is deactivated, isolating the storage nodes from the bit lines.

Steps involved in the read operation

1. Precharge both bit lines to VDD.
2. Activate the word line.
3. Access transistors conduct.
4. One bit line discharges slightly.
5. Sense amplifier detects voltage difference.
6. Output data is produced.
7. Word line is deactivated.

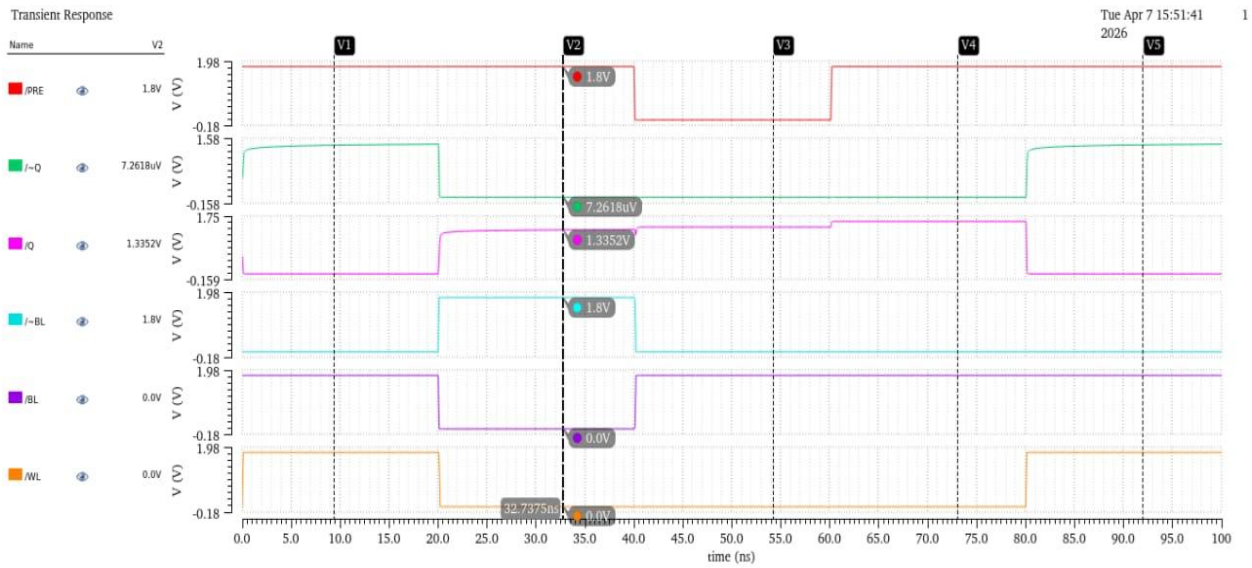


Figure 3.3: write and hold operation waveform.

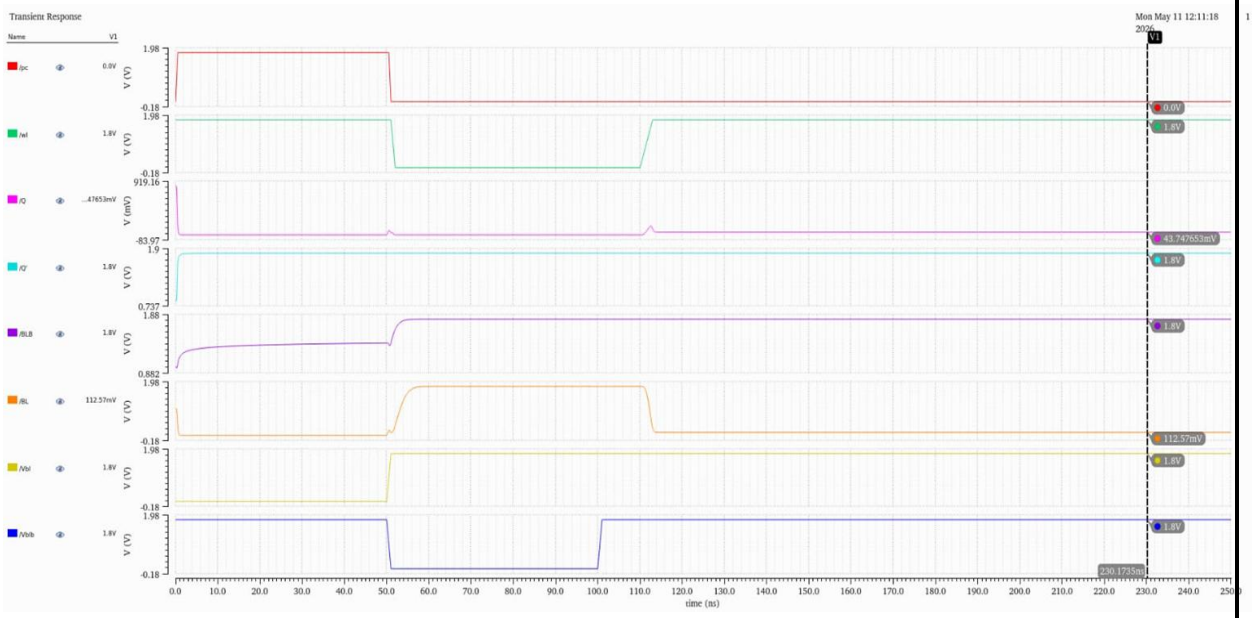


Figure 3.3: Read operation waveform.

3.4 Write Operation

The write operation changes the stored logic value inside the SRAM cell.

Before enabling the word line, the external write driver forces complementary values onto the bit lines. For example, to write logic '1', BL is driven high while BLB is driven low.

When the word line becomes active, the access transistors connect the storage nodes to the bit lines. Since the write drivers are stronger than the internal feedback of the cross-coupled inverters, the previous stored value is overwritten.

Once the desired state has been established, the word line is deactivated, and the new data remains stored through positive feedback.

Successful writing depends upon the relative strengths of the access transistors and the inverter transistors.

Steps involved in the write operation

1. Desired data applied to BL and BLB.
2. Word line activated.
3. Access transistors conduct.
4. Previous data overwritten.
5. Cross-coupled inverters settle into new state.
6. Word line deactivated.

3.5 Hold Operation

During the hold state, the SRAM cell stores data without being accessed.

The word line remains LOW, thereby turning OFF both access transistors. This disconnects the storage nodes from the bit lines, leaving the cross-coupled inverters isolated.

Because of the regenerative feedback provided by the inverters, the stored data is continuously reinforced, enabling the memory cell to retain its information indefinitely as long as power is supplied.

The hold operation is the normal operating condition of SRAM when neither reading nor writing is taking place.

3.6 Precharge Circuit

The precharge circuit prepares the bit lines before every read operation.

Since SRAM read operations depend upon detecting very small voltage differences between complementary bit lines, both BL and BLB must begin at identical voltages. The precharge circuit accomplishes this by charging both bit lines to the supply voltage before every memory access.

Typically, PMOS transistors controlled by a precharge signal connect both bit lines to VDD. In many implementations, an additional equalization transistor connects the two bit lines together during precharge to eliminate any residual voltage difference.

Without a precharge circuit, the initial voltages of the bit lines could differ, causing incorrect sensing, slower access times, and unreliable read operations.

In this project, a dedicated precharge circuit was designed and integrated with the SRAM cell to ensure proper read functionality before performing radiation fault injection experiments.

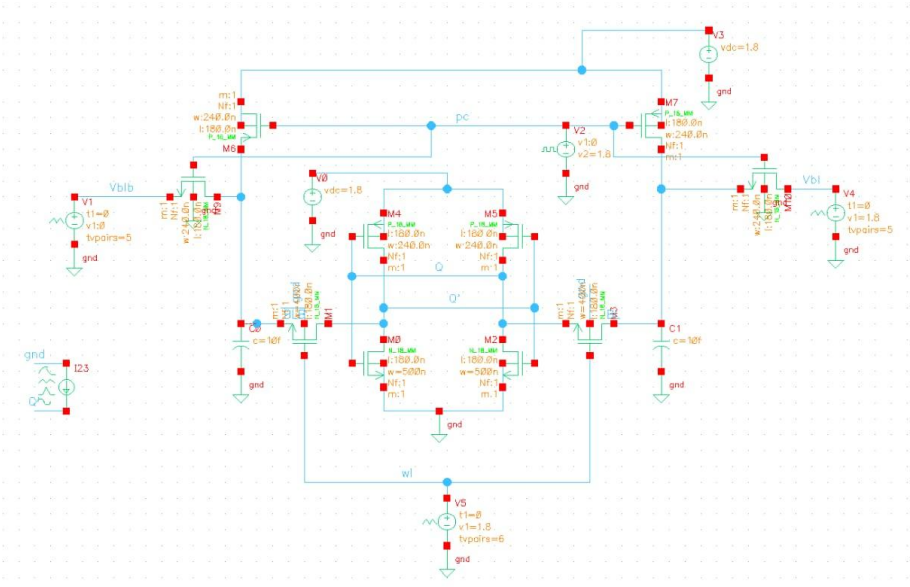


Figure 3.6: 6T SRAM with Pre-charge circuit schematic.

3.7 Charge Injection

Radiation effects were simulated by injecting a transient current pulse into one of the sensitive storage nodes of the SRAM cell.

When an energetic particle strikes the reverse-biased drain region of a MOS transistor, it generates a large number of electron-hole pairs. These carriers are rapidly collected by the electric field present within the depletion region, producing a transient current pulse.

Instead of physically exposing the circuit to radiation, Cadence simulations reproduce this phenomenon using a double exponential current source.

The injected current pulse is expressed as

$$\mathbf{I(t) = I_p(e^{(-t/\tau_2)} - e^{(-t/\tau_1)})}$$

where:

- I_p = Peak current
- τ_1 = Rise time constant
- τ_2 = Fall time constant

The rise time models the rapid generation of charge immediately after particle impact, while the fall time represents the gradual recombination and collection of charge carriers.

By varying the peak current, the amount of injected charge increases until the stored logic flips, thereby identifying the critical charge of the memory cell.

3.8 Critical Charge

Critical charge ($Q_{(}crit_{)}$) is the minimum amount of collected charge required to change the stored logic state of a memory cell.

It is one of the most important parameters used to evaluate the radiation tolerance of SRAM cells. If the deposited charge is less than the critical charge, the cell successfully restores its original state. However, when the deposited charge exceeds the critical value, the regenerative feedback of the cross-coupled inverters drives the memory into the opposite stable state, producing a Single Event Upset.

For an ideal constant current pulse, the deposited charge is given by
$\mathbf{Q = I * t}$

where

- **Q** = Collected charge (Coulombs)
- **I** = Current (Amperes)
- **t** = Pulse duration (Seconds)

For practical radiation simulations using a double exponential current pulse, the collected charge is more accurately approximated by $Q_{crit}=I_{peak}\ t_{fall}\text{-}t_{rise}$

This expression was employed in the present work to estimate the threshold charge responsible for producing Single Event Upsets.

A higher critical charge indicates better radiation tolerance because a larger amount of deposited charge is required to disturb the stored data. Conversely, advanced CMOS technologies generally exhibit smaller critical charges due to reduced node capacitance and lower supply voltages, making them more susceptible to radiation-induced soft errors.

In this project, the critical charge of the conventional 6T SRAM cell was determined by gradually increasing the magnitude of the injected current pulse until the stored logic state flipped. The same procedure was then applied to the DICE SRAM architecture, which successfully recovered from the transient disturbance, demonstrating its superior resistance to Single Event Upsets.

Chapter Summary

This chapter presented the theoretical concepts required for understanding radiation-induced failures in SRAM cells. The fundamentals of CMOS technology, the structure and operation of the conventional 6T SRAM cell, and the principles of read, write, and hold operations were discussed. The role of the precharge circuit in reliable memory access was explained, followed by the mechanism of radiation-induced charge injection using a double exponential current pulse. Finally, the concept of critical charge, which serves as the primary metric for evaluating SEU susceptibility, was introduced. These theoretical foundations provide the basis for the implementation and simulation results presented in the following chapters.

CHAPTER 4

TOOLS AND DESIGN METHODOLOGY

4.1 Software Used

The design and simulation of the radiation-hardened SRAM cell were carried out using the Cadence Virtuoso Analog Design Environment (ADE), one of the most widely used Electronic Design Automation (EDA) tools for integrated circuit design. Cadence Virtuoso provides a comprehensive platform for schematic capture, circuit simulation, waveform analysis, and verification of analog and mixed-signal circuits.

The software was used to construct the conventional 6T SRAM cell, the precharge circuit, and the Dual Interlocked Storage Cell (DICE) SRAM architecture. It also enabled transient simulations required to verify the functionality of the memory cells during read, write, and hold operations. Furthermore, the Analog Design Environment was utilized to inject transient current pulses into the storage nodes to emulate Single Event Upsets (SEUs) and analyze the radiation tolerance of the designed circuits.

The Spectre circuit simulator integrated within Cadence Virtuoso was employed to perform transient analysis. Spectre is a high-performance SPICE-compatible simulator capable of accurately modeling transistor-level circuit behavior under various operating conditions. The simulator was used to analyze voltage waveforms, current flow, and the response of the SRAM cells to radiation-induced disturbances.

The waveform results were analyzed using Cadence's waveform viewer, allowing detailed observation of voltage transitions at internal storage nodes and enabling accurate determination of the critical charge required to induce a Single Event Upset.

Overall, Cadence Virtuoso provided a complete design environment for schematic development, simulation, verification, and performance evaluation throughout this project.

4.2 Design Methodology

The project was carried out in a systematic sequence to evaluate the susceptibility of conventional SRAM cells to radiation-induced soft errors and to investigate the effectiveness of a radiation-hardened architecture.

Initially, the functional requirements of the project were identified, including the design of a conventional 6T SRAM cell capable of performing reliable read, write, and hold operations. Based on these requirements, the complete transistor-level schematic of the SRAM cell was developed using CMOS devices available in the selected technology library.

Once the SRAM cell was designed, transient simulations were performed to verify its basic functionality. Separate simulations were carried out for write, read, and hold operations to ensure that the memory cell operated correctly under normal conditions. A precharge circuit was then designed and integrated with the SRAM cell to charge the bit lines before each read operation, thereby ensuring reliable sensing of stored data.

After validating the normal operation of the memory cell, radiation effects were simulated by injecting a double exponential transient current pulse into one of the sensitive storage nodes. The injected current was gradually increased until the stored logic state flipped, allowing the critical charge threshold of the conventional SRAM cell to be determined.

Following the analysis of the conventional SRAM, a Dual Interlocked Storage Cell (DICE) SRAM architecture was designed using Cadence Virtuoso. Similar functional verification was performed to confirm correct read, write, and hold operations. The same transient current pulse used for the conventional SRAM was then applied to the DICE SRAM under identical simulation conditions.

Finally, the simulation results of both memory architectures were compared with respect to their ability to withstand radiation-induced disturbances. The comparison focused on their susceptibility to Single Event Upsets, data recovery characteristics, and overall radiation tolerance. This systematic methodology enabled a comprehensive evaluation of the effectiveness of the DICE architecture in mitigating soft errors.

4.3 Technology Used

The circuits developed in this project were implemented using the 180 nm CMOS technology node. This technology provides a balanced combination of design simplicity, simulation accuracy, and device reliability, making it suitable for academic research and educational VLSI projects.

In a 180 nm CMOS process, the minimum transistor channel length is approximately 180 nanometers. The technology includes both NMOS and PMOS transistors with well-characterized electrical models, allowing accurate simulation of analog and digital circuits. Compared with more advanced technology nodes, the 180 nm process exhibits relatively larger transistor dimensions and higher node capacitances, which simplify circuit analysis while still demonstrating the fundamental behavior of radiation-induced effects.

The technology library supplies transistor models, design rules, and electrical parameters required for schematic design and circuit simulation. These models account for device characteristics such as threshold voltage, channel dimensions, parasitic capacitances, leakage currents, and switching behavior, enabling realistic simulation of SRAM operation.

A supply voltage compatible with the 180 nm technology was used throughout the simulations. The selected technology also supports reliable transient analysis, making it suitable for investigating Single Event Upsets through current pulse injection. Although modern commercial processors are fabricated using much smaller technology nodes, the 180 nm process remains widely used for academic research because it provides stable device models, reduced simulation complexity, and a clear understanding of transistor-level circuit behavior.

The use of the 180 nm CMOS technology in this project allowed accurate evaluation of the conventional 6T SRAM cell and the DICE SRAM architecture under identical operating conditions. This ensured that the comparison between the two memory structures was based solely on architectural differences rather than variations in fabrication technology.

Chapter Summary

This chapter described the software tools and design methodology adopted for the implementation of the project. Cadence Virtuoso and the Spectre simulator were used to design, simulate, and analyze both the conventional 6T SRAM cell and the DICE SRAM architecture. A structured methodology involving schematic design, functional verification, transient current injection, and comparative analysis was followed throughout the project. The circuits were implemented using the 180 nm CMOS technology node, providing a reliable platform for investigating radiation-induced Single Event Upsets and evaluating the effectiveness of radiation-hardened memory design techniques.

CHAPTER 5

IMPLEMENTATION OF CONVENTIONAL 6T SRAM

5.1 Schematic of the 6T SRAM Cell

The first stage of this project involved the design and implementation of a conventional 6-Transistor (6T) Static Random Access Memory (SRAM) cell using the Cadence Virtuoso design environment. The objective of this design was to establish a baseline memory architecture before studying the effects of radiation-induced Single Event Upsets and comparing its performance with the radiation-hardened DICE SRAM architecture.

The conventional SRAM cell consists of six MOS transistors arranged as two cross-coupled CMOS inverters and two access transistors. The cross-coupled inverters form a bistable latch capable of storing one bit of information, while the access transistors provide connectivity between the storage nodes and the complementary bit lines during read and write operations.

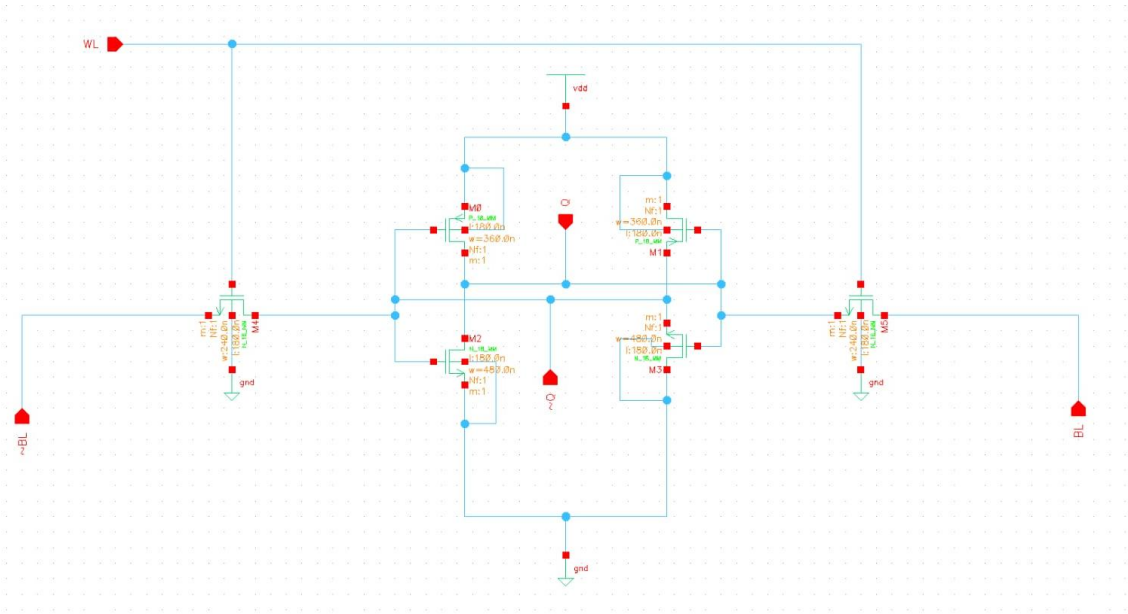


Figure 5.1: Schematic of the conventional 6T SRAM cell.

Transistor Description

The functionality of each transistor in the SRAM cell is described below.

Pull-Up PMOS Transistors (P1 and P2)

The two PMOS transistors form the pull-up network of the SRAM cell.

Their functions are:

- Connect the storage nodes to VDD.
- Maintain the stored logic '1'.
- Restore the storage node voltage after transient disturbances.
- Participate in the positive feedback mechanism of the cross-coupled inverters.

Whenever one storage node stores logic '0', the corresponding PMOS transistor turns ON, charging the opposite storage node to VDD and maintaining the complementary logic state.

Pull-Down NMOS Transistors (N1 and N2)

The two NMOS transistors form the pull-down network.

Their functions include:

- Connecting storage nodes to ground.
- Maintaining logic '0'.
- Providing a discharge path during read operations.

- Assisting in switching during write operations.

The pull-down transistors are generally designed stronger than the access transistors to prevent accidental flipping of the stored data during read operations.

Access NMOS Transistors (A1 and A2)

The access transistors connect the internal storage nodes to the complementary bit lines.

Their functions are:

- Controlled by the Word Line (WL).
- Enable data transfer during read operations.
- Enable data writing from the bit lines.
- Isolate the memory cell during hold operation.

When the word line remains LOW, both access transistors remain OFF, electrically isolating the storage nodes from the external circuitry.

Working Principle

The two CMOS inverters are connected in a feedback configuration. Due to regenerative positive feedback, the circuit possesses two stable operating states:

State 1

Q = 1

QB = 0

State 2

Q = 0

QB = 1

Once either state is established, the cross-coupled structure continuously reinforces the stored value until power is removed or new data is written.

5.2 Functional Verification of the SRAM Cell

Before evaluating radiation tolerance, it was necessary to verify that the SRAM cell operated correctly under normal operating conditions.

Three functional simulations were performed:

- Read Operation
- Write Operation
- Hold Operation

Each simulation was carried out using transient analysis in Cadence Spectre.

5.3 Read Operation

The read operation retrieves the stored information without modifying the stored data.

Prior to enabling the word line, both bit lines were precharged to the supply voltage using the precharge circuit. This ensured identical initial conditions for both bit lines.

When the word line became HIGH, both access transistors turned ON, connecting the storage nodes to BL and BLB.

Suppose the stored data is

Q = 1

QB = 0

Since QB stores logic '0', its pull-down NMOS transistor provides a discharge path for BLB. Consequently,

- BL remains approximately equal to VDD.
- BLB experiences a slight voltage drop.

Although the voltage difference is relatively small, it is sufficient for the sense amplifier to distinguish the stored logic value.

After completion of the read cycle, the word line returns LOW, disconnecting the SRAM cell from the bit lines.

Waveform Analysis

The waveform demonstrates that:

- Word Line transitions HIGH during the read interval.
- Both access transistors conduct.
- One bit line remains near VDD.
- The complementary bit line discharges slightly.
- Internal storage nodes remain unchanged.
- Stored data is preserved after the read cycle.

This confirms that the read operation is non-destructive.

5.4 Write Operation

The write operation modifies the logic state stored within the SRAM cell.

To write new data, complementary voltages are first applied to the bit lines.

For example,

To write logic '1'

BL = VDD

BLB = 0

When the word line is activated, the access transistors connect the bit lines directly to the storage nodes.

The write drivers force the desired voltage levels onto the storage nodes. Once the internal node voltage crosses the inverter switching threshold, regenerative feedback rapidly switches the cross-coupled inverters into the new stable state.

Following completion of the write operation, the word line is deactivated.

The newly written value is retained indefinitely until another write operation occurs.

Waveform Analysis

The simulation waveform shows that:

- Complementary data appears on BL and BLB.
- Word Line becomes HIGH.
- Internal storage nodes begin switching.
- Positive feedback accelerates the transition.
- Q and QB exchange logic values.
- The new data remains stable after WL becomes LOW.

The simulation verifies successful writing of the desired logic value.

5.5 Hold Operation

The hold operation represents the idle condition of the SRAM cell.

During this mode,

- Word Line remains LOW.
- Access transistors remain OFF.
- Storage nodes are isolated from BL and BLB.

The cross-coupled CMOS inverters continuously regenerate the stored data through positive feedback.

Since there is no conductive path to the external circuitry, the stored information remains unchanged.

Waveform Analysis

The simulation demonstrates:

- Word Line remains LOW.
- Storage nodes maintain constant voltages.
- No switching activity occurs.
- No data corruption is observed.
- The SRAM successfully retains the stored information.

The waveform confirms the stability of the memory cell during standby operation.

5.6 Precharge Circuit

The precharge circuit is an essential component of SRAM design because reliable read operations depend on both bit lines starting from identical voltage levels.

Without precharging, one bit line may already be partially discharged due to the previous memory access, resulting in inaccurate sensing and increased read delay.

Principle of Operation

The implemented precharge circuit consists of PMOS transistors controlled by a precharge enable signal.

When the precharge signal is LOW,

- PMOS transistors turn ON.
- BL is charged to VDD.
- BLB is charged to VDD.
- Both bit lines reach identical voltages.

When the precharge signal becomes HIGH,

- PMOS transistors turn OFF.
- Bit lines become electrically isolated.
- Read operation begins.

The precharge circuit ensures that the differential voltage generated during the read operation depends solely on the stored data and not on any residual charge remaining from previous operations.

Importance of the Precharge Circuit

The precharge circuit offers several advantages:

- Establishes equal initial voltage on both bit lines.
- Improves sensing accuracy.
- Reduces read access time.
- Minimizes read errors.
- Improves overall memory stability.
- Ensures repeatable simulation results.

Without precharging, the sense amplifier may incorrectly interpret stored data due to unequal initial conditions on BL and BLB.

Simulation of the Precharge Circuit

The designed precharge circuit was simulated independently before integrating it with the SRAM cell.

The transient simulation verified that:

- Both bit lines charged completely to VDD.
- Charging occurred rapidly.
- Both bit lines exhibited identical voltages before each read cycle.
- No residual voltage difference remained after precharging.

After successful verification, the precharge circuit was connected to the 6T SRAM cell and used throughout all subsequent read and radiation simulations.

Waveform Analysis

The waveform clearly indicates:

- Activation of the precharge signal.
- Simultaneous charging of BL and BLB.
- Equalization of both bit lines.
- Stable voltage prior to word line activation.
- Successful transition into the read cycle.

The simulation confirms that the implemented precharge circuit performs its intended function effectively and provides the necessary initial conditions for reliable SRAM operation.

Chapter Summary

This chapter presented the complete implementation of the conventional 6T SRAM cell using the Cadence Virtuoso design environment. The transistor-level schematic was developed and the functionality of each transistor was explained in detail. The designed SRAM cell was successfully verified through transient simulations of read, write, and hold operations. A dedicated precharge circuit was also implemented and validated to ensure reliable bit-line initialization before every read operation. The successful completion of these functional simulations established a reliable baseline for the radiation fault injection experiments discussed in the next chapter.

CHAPTER 6

SINGLE EVENT UPSET (SEU) ANALYSIS

Single Event Upset (SEU) is one of the most significant reliability concerns in modern CMOS memories. As transistor dimensions continue to shrink, the amount of charge stored at each memory node decreases, making SRAM cells increasingly susceptible to radiation-induced soft errors. High-energy particles such as alpha particles, heavy ions, neutrons, or cosmic rays generate electron-hole pairs while passing through the silicon substrate. The collected charge appears as a transient current pulse at the sensitive storage node. If this injected charge exceeds the critical charge of the cell, the stored logic value changes, resulting in a bit flip.

In this work, SEU analysis is carried out using transient current injection. Current pulses of different magnitudes are injected into the sensitive storage nodes while observing the voltage response of the SRAM cell. The objective is to determine the threshold current required for upsetting the stored data and to calculate the critical charge (Qcrit) of the proposed radiation-hardened SRAM cell.

6.1 Current Pulse Injection

To emulate the effect of an energetic particle strike, a double-exponential current pulse is injected at the storage node of the SRAM cell. The double-exponential waveform closely represents the transient current generated by charge collection inside semiconductor devices after ionizing radiation.

The injected current pulse is mathematically represented as

$$I(t) = I_{\{peak\}}\left(e^{\left\{-\frac{t}{\tau_{\{fall\}}}\right\}} - e^{\left\{-\frac{t}{\tau_{\{rise\}}}\right\}}\right)$$

where

- I_{peak} = Peak injected current

  tau_{rise} = Rise time constant

- tau_{fall} = Fall time constant

Initially the current rises rapidly because of the instantaneous generation of charge carriers. The current then decays exponentially as the excess carriers recombine and diffuse away from the strike location.

The parameters varied during simulation are:

- Peak current
- Pulse duration
- Injection node
- Stored logic state

For each simulation, the storage node voltages were monitored to determine whether the SRAM cell retained its data or experienced a bit flip.

The current waveform demonstrates the rapid rise followed by exponential decay characteristic of a radiation-induced transient. Such a pulse accurately models the electrical effect of particle strikes in deep submicron CMOS technologies.

6.2 Charge Injection Analysis

When the transient current pulse is applied to one of the storage nodes, charge is deposited on the internal capacitances of the SRAM cell. This temporarily disturbs the node voltage.

If the deposited charge is small, the positive feedback of the cross-coupled inverters restores the original logic state. However, when the injected charge becomes sufficiently large, the disturbed node crosses the inverter switching threshold. The regenerative action of the cross-coupled pair then forces the complementary node to switch, permanently changing the stored data.

Two simulations were performed:

- Stored value = 0
- Stored value = 1

Case 1: Stored Logic '0'



Figure 6.1: Transient Response for Stored Logic '0'

Initially, node **Q** stores logic '0' while node $\bar{\mathbf{Q}}$ stores logic '1'. At approximately **65 ns**, a current pulse is injected into the sensitive node.

The waveform shows that:

- The struck node voltage begins to increase.

- The complementary node starts decreasing simultaneously.
- The disturbance grows because of the regenerative feedback of the cross-coupled inverters.
- Once the switching threshold is exceeded, both nodes interchange their logic states.

The final stable voltages indicate that the stored bit has flipped successfully from '0' to '1'. This confirms that the injected charge exceeded the critical charge of the SRAM cell.

Case 2: Stored Logic '1'

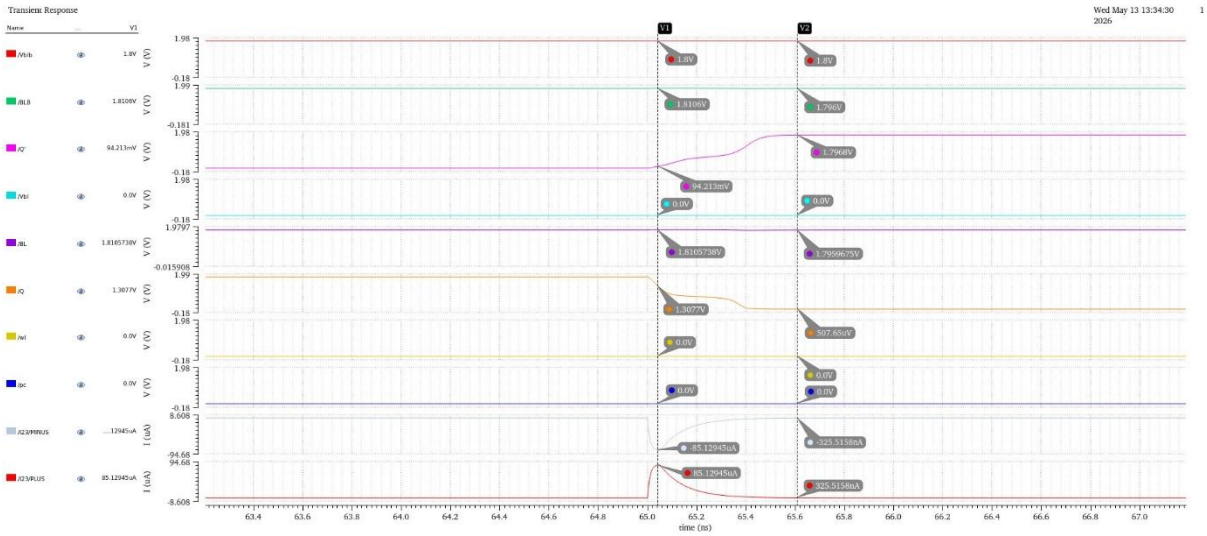


Figure 6.2: Transient Response for Stored Logic '1'

In the second experiment, the SRAM initially stores logic '1'. A transient current pulse is injected at the complementary sensitive node.

The waveform indicates:

- A temporary voltage disturbance immediately after current injection.
- The storage node voltage experiences a transient dip.
- The complementary node exhibits a short voltage spike.
- After the transient disappears, both nodes recover their original logic levels.

This demonstrates that the injected charge was insufficient to overcome the restoring capability of the cross-coupled inverters. Therefore, no permanent upset occurred.

The comparison of both simulations clearly shows that the response depends not only on the injected current but also on the initial stored value and the sensitive node selected for injection.

6.3 Threshold Current Determination

One of the most important objectives of SEU analysis is to determine the minimum injected current capable of causing a bit flip. This value is known as the **threshold current**.

The threshold current was obtained through an iterative simulation procedure.

The procedure followed is:

1. A small current pulse was injected into the storage node.
2. The transient response was observed.
3. If the SRAM retained its stored value, the peak current was increased.
4. The process was repeated until a permanent bit flip occurred.
5. The smallest current producing a successful upset was recorded as the threshold current.

The simulation outcomes are summarized in the following format.

Peak Current	Pulse Width	Injected Charge	Cell Response
Low	Fixed	Low	No Bit Flip
Medium	Fixed	Moderate	Stable
Threshold	Fixed	Critical	Bit Flip
High	Fixed	Large	Immediate Bit Flip

From the simulations, three operating regions were identified.

Region 1: No Bit Flip

The injected charge is lower than the restoring capability of the SRAM cell.

Characteristics:

- Temporary voltage disturbance
- Complete recovery
- Stored data retained

Region 2: Threshold Region

The injected current is just sufficient to force one storage node across the inverter trip point.

Characteristics:

- Marginal stability
- Slow recovery

- Beginning of regenerative switching

This region defines the critical operating point of the SRAM.

Region 3: Bit Flip Region

When the injected current exceeds the threshold,

- Regenerative feedback dominates.
- Both storage nodes interchange their logic values.
- The SRAM settles to the opposite stable state.

The threshold current therefore represents the minimum particle-induced disturbance capable of causing a soft error.

6.4 Critical Charge (Qcrit) Calculation

The critical charge ((Q_{crit})) is defined as the minimum amount of collected charge required to change the stored logic state of the SRAM cell permanently.

For the double exponential current model,

$$Q_{\{crit\}} = I_{\{peak\}}(\tau_{\{fall\}} - \tau_{\{rise\}})$$

where

- $(I_{\{peak\}})$= Threshold peak current
- $(\tau_{\{fall\}})$= Current decay constant
- $(\tau_{\{rise\}})$= Current rise constant

Using the threshold current obtained from transient simulations, the critical charge was calculated for both storage nodes.

Stored Logic '0'

Node	Threshold Current (µA)	Critical Charge (fC)
Q	318.6	28.67
$\bar{Q}$	240.31	21.63

The larger critical charge required for node Q indicates higher immunity against radiation-induced upsets when storing logic '0'.

Stored Logic '1'

Node	Threshold Current (μA)	Critical Charge (fC)
Q	86.65	7.80
$\bar{Q}$	98.85	8.90

Compared to the stored logic '0' case, significantly smaller critical charges are sufficient to upset the cell when storing logic '1'. This indicates that the cell exhibits asymmetric SEU sensitivity, which is common in SRAM architectures because of differences in transistor operating conditions and node capacitances.

Charge versus Stability

The relationship between injected charge and SRAM stability is illustrated conceptually below.

Injected Charge	SRAM Behaviour
$(Q < Q_{\{crit\}})$	Stable operation
$(Q \approx Q_{\{crit\}})$	Marginal stability
$(Q > Q_{\{crit\}})$	Permanent bit flip

As the injected charge increases, the stability margin of the SRAM decreases. The regenerative feedback initially suppresses small disturbances, but once the deposited charge exceeds the critical charge, the feedback mechanism reinforces the disturbance and forces the cell into the opposite stable state.

Therefore, increasing the critical charge improves radiation hardness and enhances the reliability of SRAM memories operating in radiation-prone environments such as aerospace, satellites, and nuclear electronics.

Chapter Summary

This chapter presented the Single Event Upset analysis of the proposed SRAM cell using transient current injection. A double-exponential current pulse was used to emulate radiation-induced particle strikes. The voltage responses for stored logic '0' and logic '1' were analyzed to study upset behavior. The minimum threshold current required to cause a bit flip was determined through iterative simulations, and the corresponding critical charge values were calculated. The obtained Q_{crit} values quantify the radiation tolerance of the SRAM cell, demonstrating that larger critical charge corresponds to greater resistance against soft errors and improved overall stability.

CHAPTER 7

IMPLEMENTATION OF DICE SRAM

7.1 Theory of DICE SRAM

As CMOS technology continues to scale toward smaller technology nodes, memory cells become increasingly vulnerable to radiation-induced soft errors. The reduction in transistor dimensions, supply voltage, and node capacitance significantly lowers the amount of charge stored within the memory cell. Consequently, even a single energetic particle striking a sensitive storage node may deposit sufficient charge to alter the stored logic state, resulting in a Single Event Upset (SEU).

To improve the reliability of SRAM cells operating in radiation-prone environments, Radiation-Hardened-by-Design (RHBD) techniques have been developed. Among these techniques, the **Dual Interlocked Storage Cell (DICE)** architecture is one of the most effective because it provides fault tolerance through circuit redundancy rather than relying on specialized fabrication processes.

Unlike the conventional 6T SRAM cell, which stores one bit using two cross-coupled inverters, the DICE SRAM stores the same information using **four mutually interlocked storage nodes**. The information is therefore represented redundantly, ensuring that a disturbance occurring at one node does not immediately propagate to the remaining storage nodes.

The DICE architecture employs twelve MOS transistors consisting of four interconnected storage elements and two access transistors. The four storage nodes are generally represented as **A, B, C, and D**. During normal operation, two nodes maintain logic '1' while the remaining two maintain logic '0'. Since each node is reinforced by multiple feedback paths, disturbing one node alone is insufficient to permanently alter the stored information.

This redundant storage mechanism forms the basis of the self-correcting capability of the DICE SRAM. When one storage node experiences a transient disturbance, the remaining nodes continue preserving the original logic state and automatically restore the disturbed node after the transient disappears. Consequently, the DICE architecture exhibits significantly greater immunity to radiation-induced Single Event Upsets than the conventional 6T SRAM.

7.2 Implementation of the DICE SRAM Cell

After successfully implementing and verifying the conventional 6T SRAM cell, the DICE SRAM architecture was designed using the Cadence Virtuoso Analog Design Environment.

The circuit was implemented using the same **180 nm CMOS technology** employed for the conventional SRAM design. Using identical technology parameters ensured that the comparison between the two architectures depended solely on their circuit structures rather than fabrication differences.

The DICE SRAM consists of twelve MOS transistors arranged to create four interlocked storage nodes. Two access transistors controlled by the Word Line connect the storage

nodes to the complementary bit lines during read and write operations. Internally, the storage nodes are connected in such a manner that each node is continuously reinforced by the remaining nodes through positive feedback.

Following schematic completion, transistor connectivity was verified and transient simulations were performed to ensure correct operation before radiation testing.

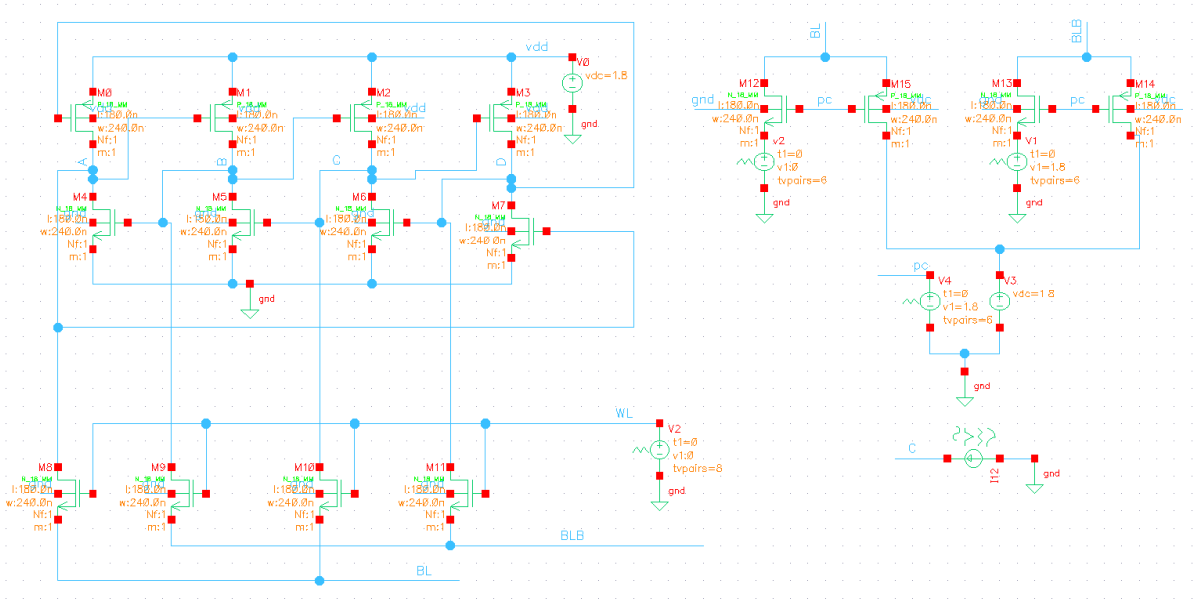


Figure 7.1: Transistor-level schematic of the implemented DICE SRAM cell.

The implemented schematic successfully demonstrated the expected interlocked storage structure required for radiation-hardened operation.

7.3 Functional Verification and SEU Simulation

Before investigating the radiation tolerance of the DICE SRAM cell, its normal functionality was verified through transient simulations. The simulations included read, write, and hold operations to confirm that the memory cell performed identically to a conventional SRAM under normal operating conditions.

During the **write operation**, complementary logic values were applied to the bit lines while the Word Line was asserted. The access transistors connected the bit lines to the internal storage nodes, allowing the desired logic value to be stored. Once the Word Line returned LOW, the interlocked feedback network maintained the stored information.

The **read operation** was performed after precharging both bit lines to the supply voltage. When the Word Line became active, one of the bit lines experienced a slight discharge depending on the stored logic value. The differential voltage generated between the two bit lines was sufficient to determine the stored data without disturbing the internal storage nodes.

During the **hold operation**, the Word Line remained LOW, isolating the storage nodes from the external circuitry. The regenerative feedback among the four storage nodes continuously reinforced the stored information, ensuring stable data retention throughout the simulation interval.

After confirming the correct functionality of the memory cell, radiation effects were simulated by injecting a **double exponential transient current pulse** into one of the sensitive storage nodes. The same current pulse parameters used for the conventional 6T SRAM analysis were employed to enable a direct comparison between the two architectures.

Immediately after current injection, the struck node exhibited a temporary voltage disturbance. However, unlike the conventional SRAM cell, this disturbance did not propagate to the remaining storage nodes. Instead, the unaffected nodes maintained their original logic values and gradually restored the disturbed node through the interlocked feedback network.

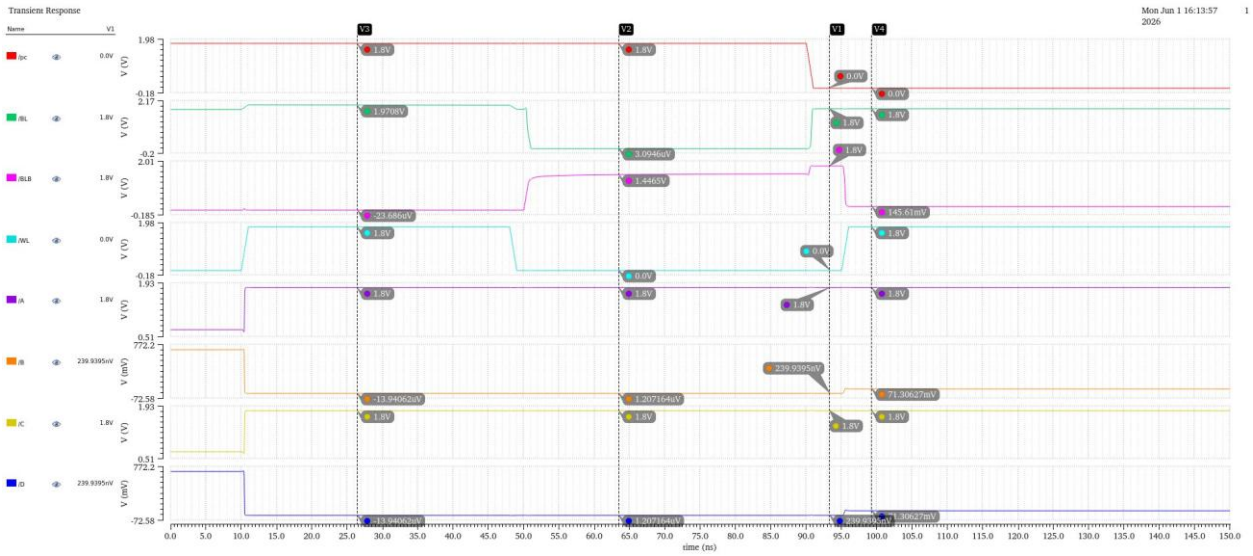


Figure 7.2: Combined transient simulation waveforms showing the read operation, write operation, hold operation.

The waveform demonstrates the complete operation of the DICE SRAM cell under both normal and radiation-induced conditions. During the read operation, the stored data is accessed correctly without altering the memory contents. The write operation successfully updates the stored value, while the hold operation confirms stable data retention. Following the injection of the transient current pulse, one storage node experiences a temporary voltage deviation. Nevertheless, the remaining storage nodes continue maintaining the correct logic state, preventing the disturbance from propagating throughout the memory cell.

As the transient current pulse decays, the regenerative feedback among the four storage nodes automatically restores the disturbed node to its original voltage level. The final steady-state voltages are identical to those observed before the radiation event, confirming that no permanent bit flip has occurred.

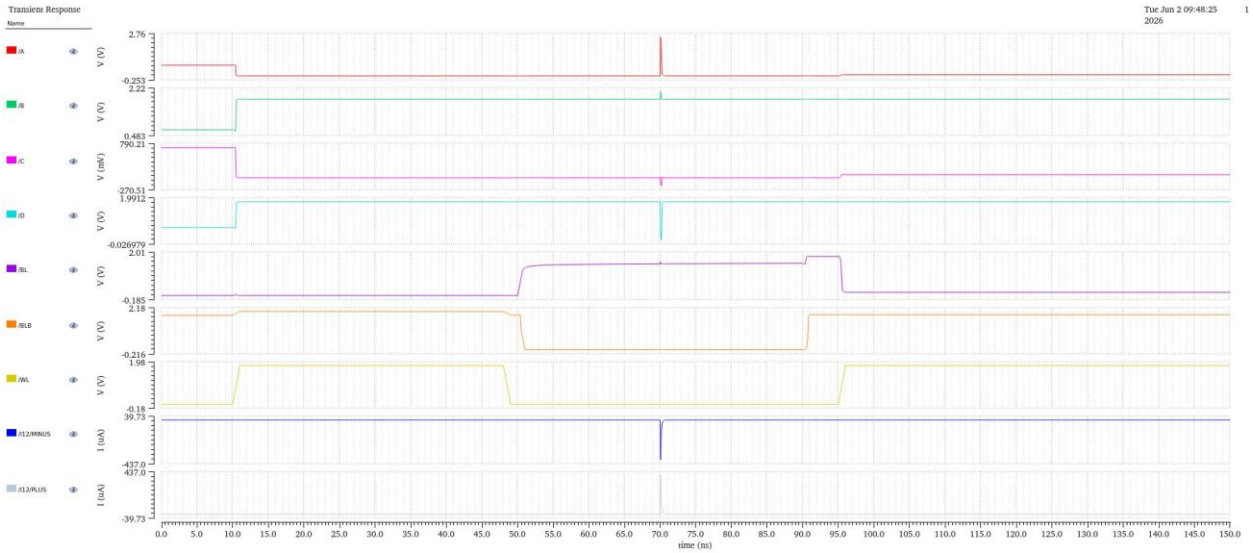


Figure 7.3: Combined waveforms for current pulse injection, and the storage node voltages before current injection, during the transient disturbance, and after automatic recovery when Q=0

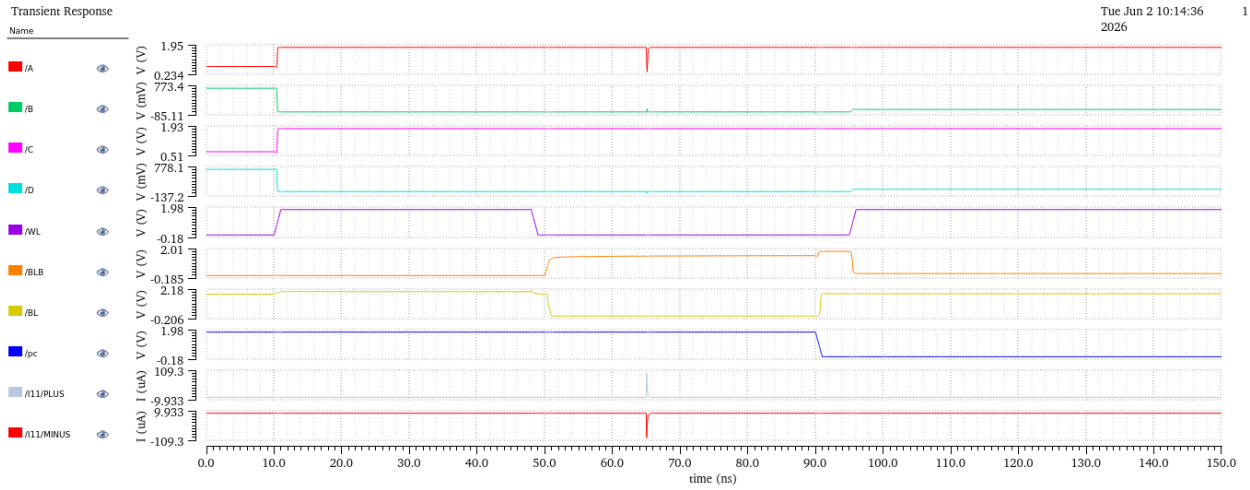


Figure 7.4: Combined waveforms for current pulse injection, and the storage node voltages before current injection, during the transient disturbance, and after automatic recovery when Q=1

7.4 Recovery Mechanism of the DICE SRAM

The most significant contribution of this project is the demonstration of the self-recovery capability of the DICE SRAM architecture following a simulated Single Event Upset.

Initially, before current injection, all four storage nodes remain in their intended logic states and the memory cell stores valid data. The interlocked feedback network maintains a stable equilibrium, ensuring reliable memory operation.

When the transient current pulse is injected into one of the sensitive nodes, the voltage at the struck node deviates temporarily due to the deposited charge. In a conventional 6T SRAM cell, such a disturbance would propagate through the cross-coupled inverters and permanently invert the stored logic state once the injected charge exceeded the critical charge threshold.

The behavior of the DICE SRAM differs fundamentally because the remaining storage nodes continue preserving the correct logic value. Since three unaffected nodes simultaneously reinforce the original data, the disturbed node is unable to force the entire memory cell into the opposite stable state.

Once the transient current pulse disappears, the positive feedback provided by the remaining storage nodes gradually restores the disturbed node to its original voltage level. Consequently, the memory cell returns to exactly the same stable condition that existed before the radiation event.

This recovery process demonstrates that the DICE architecture is capable of tolerating Single Event Upsets affecting individual storage nodes without requiring any external correction circuitry. The redundant storage mechanism significantly increases the radiation tolerance of the memory cell and improves its suitability for applications requiring high reliability.

The simulation results obtained in this project clearly illustrate this behavior. While the conventional 6T SRAM experienced a permanent logic inversion once the injected charge exceeded its critical value, the DICE SRAM successfully recovered after the transient disturbance. This confirms the effectiveness of the Dual Interlocked Storage Cell architecture as a Radiation-Hardened-by-Design technique for mitigating Single Event Upsets.

Chapter Summary

This chapter presented the implementation and simulation of the DICE SRAM architecture using Cadence Virtuoso in 180 nm CMOS technology. The transistor-level schematic of the DICE SRAM was successfully developed and verified. Functional simulations confirmed correct read, write, and hold

operations under normal operating conditions. A double exponential current pulse was subsequently injected into one of the sensitive storage nodes to emulate a radiation-induced particle strike. The simulation results demonstrated that although the injected current produced a temporary voltage disturbance, the interlocked storage nodes automatically restored the original logic state after the transient pulse disappeared. Unlike the conventional 6T SRAM, no permanent bit flip was observed. These results validate the effectiveness of the DICE SRAM architecture as a Radiation-Hardened-by-Design solution for improving memory reliability in aerospace, satellite, automotive, medical, and other safety-critical electronic systems.

CHAPTER 8

RESULTS AND DISCUSSION

8.1 Overview

The primary objective of this project was to investigate the susceptibility of a conventional 6T SRAM cell to radiation-induced Single Event Upsets (SEUs) and to evaluate the effectiveness of the Dual Interlocked Storage Cell (DICE) architecture in mitigating these soft errors. Both memory cells were designed and simulated using the Cadence Virtuoso design environment in 180 nm CMOS technology under identical operating conditions. The conventional 6T SRAM and the DICE SRAM were first verified through read, write, and hold operations. Subsequently, transient current injection was performed to emulate the effect of energetic particle strikes and evaluate the radiation tolerance of each architecture.

The simulation results demonstrate that both memory architectures operate correctly under normal conditions. However, significant differences become evident when the memory cells are subjected to transient current injection. The conventional SRAM experiences permanent data corruption once the injected charge exceeds the critical threshold, whereas the DICE SRAM successfully restores the original data after the transient disturbance. These observations confirm that the DICE architecture provides significantly improved resistance to radiation-induced soft errors.

8.2 Functional Verification

Before analyzing radiation tolerance, the functional correctness of both SRAM architectures was verified through transient simulations. Read, write, and hold operations were successfully performed for both the conventional 6T SRAM and the DICE SRAM. During the write operation, the desired logic values were correctly stored within the memory cells. During the read operation, the stored information was retrieved without disturbing the internal storage nodes. The hold operation further demonstrated that both memory cells retained the stored data correctly when isolated from the bit lines.

These successful simulations confirmed that the circuits were functioning as intended under normal operating conditions. Therefore, any differences observed during radiation testing can be attributed solely to the differences in memory architecture rather than design errors or simulation inaccuracies.

8.3 Single Event Upset Analysis

Radiation effects were simulated by injecting a double exponential transient current pulse into the sensitive storage nodes of the SRAM cells. The injected pulse represents the electrical effect produced when an energetic particle passes through silicon and generates electron-hole pairs within the depletion region of a MOS transistor.

The injected current was gradually increased until the minimum current capable of producing a bit flip was identified. This threshold current was then used to calculate the corresponding critical charge of the conventional SRAM cell.

The transient simulations clearly showed that the conventional 6T SRAM is highly sensitive to radiation-induced disturbances. When the injected current remained below the critical threshold, the storage nodes experienced only temporary voltage deviations and eventually returned to their original logic values. However, once the injected current exceeded the critical charge threshold, the regenerative feedback of the cross-coupled inverters forced the memory cell into the opposite stable state, resulting in a permanent Single Event Upset.

The DICE SRAM responded differently under identical test conditions. Although one of its storage nodes experienced a temporary voltage disturbance following current injection, the remaining storage nodes preserved the correct logic state. As the transient pulse decayed, the disturbed node gradually returned to its original voltage level without causing any permanent change in the stored data. This behavior demonstrates the superior radiation tolerance of the DICE architecture.

8.4 Discussion of Results

The results obtained from the transient simulations clearly demonstrate the limitations of the conventional 6T SRAM architecture when operating in radiation-prone environments. The conventional SRAM relies on two cross-coupled inverters to store a single bit of information. Consequently, the stored data depends entirely on the stability of two internal storage nodes. When one of these nodes experiences a sufficiently large transient disturbance, the positive feedback mechanism of the cross-coupled inverters amplifies the disturbance and forces the memory cell into the opposite stable state. Once this transition occurs, the original data is permanently lost until rewritten.

In contrast, the DICE SRAM architecture distributes the stored information among four interlocked storage nodes rather than two. This redundancy fundamentally changes the response of the memory cell to radiation-induced disturbances. Instead of relying on a single complementary node to maintain the stored information, each storage node is continuously supported by multiple feedback paths provided by the remaining nodes.

When a transient current pulse disturbs one storage node, the unaffected nodes continue maintaining the original logic state. Because the majority of the storage network remains unchanged, the disturbed node cannot successfully force the entire memory cell into an incorrect state. As the transient current disappears, the regenerative feedback provided by the unaffected nodes gradually restores the disturbed node to its original voltage level. Therefore, the disturbance remains temporary and no permanent data corruption occurs.

This automatic recovery mechanism is the defining characteristic of the DICE architecture. Unlike conventional error correction techniques that detect and correct errors after they occur, the DICE SRAM inherently prevents the propagation of transient disturbances through its redundant storage mechanism. As a result, it significantly improves the reliability of SRAM cells without requiring continuous monitoring or additional correction circuitry.

The simulation results obtained in this project strongly support this behavior. The conventional SRAM exhibited permanent bit flips once the injected current exceeded its critical threshold, whereas the DICE SRAM consistently recovered after transient current injection under identical simulation conditions. This confirms that architectural redundancy is an effective Radiation-Hardened-by-Design technique for mitigating Single Event Upsets.

8.5 Radiation Tolerance of DICE SRAM

The improved radiation tolerance of the DICE SRAM can be attributed to three important characteristics.

The first characteristic is **node redundancy**. Unlike the conventional SRAM, which stores information using only two storage nodes, the DICE architecture distributes the stored information among four interconnected nodes. This redundancy ensures that the failure of a single node does not immediately affect the remaining storage nodes.

The second characteristic is the **interlocked feedback network**. Every storage node is continuously reinforced by the remaining nodes through multiple feedback paths. Consequently, a transient voltage disturbance at one node is counteracted by the correct logic values stored in the unaffected nodes. This regenerative action gradually restores the disturbed node after the transient pulse disappears.

The third characteristic is the **higher effective radiation tolerance** achieved through the circuit architecture itself. Since a single energetic particle generally affects only one sensitive node, the remaining nodes preserve the original data and prevent a permanent logic inversion. Only simultaneous disturbances affecting multiple storage nodes could potentially produce a successful upset, making the probability of failure significantly lower than that of a conventional 6T SRAM cell.

The simulations performed in this work clearly demonstrate these advantages. The DICE SRAM maintained correct operation during read, write, and hold operations while exhibiting excellent

resistance to radiation-induced transient disturbances. The observed self-recovery capability confirms its suitability for applications operating in harsh radiation environments.

8.6 Applications of Radiation-Hardened SRAM

Radiation-hardened memory architectures are increasingly important in modern electronic systems where reliability is critical. Satellites and spacecraft operate continuously under intense cosmic radiation and therefore require memory systems capable of resisting Single Event Upsets. Similar requirements exist in aerospace electronics, military communication systems, and nuclear instrumentation.

In the automotive industry, advanced driver assistance systems and autonomous vehicles rely on dependable electronic control units that must continue operating correctly even under transient disturbances. Medical devices such as implantable electronics, diagnostic instruments, and life-support systems also demand highly reliable memory architectures because incorrect data storage may directly affect patient safety.

Industrial automation, defense electronics, avionics, and high-performance computing systems similarly benefit from radiation-hardened SRAM architectures. The DICE SRAM presented in this project provides an effective solution for these applications by improving memory reliability through architectural redundancy rather than relying solely on specialized fabrication technologies.

Chapter Summary

This chapter presented the results obtained from the simulation and comparative analysis of the conventional 6T SRAM and the DICE SRAM architectures. Both memory cells successfully performed normal read, write, and hold operations under identical operating conditions. However, their responses to radiation-induced transient current injection differed significantly. The conventional SRAM experienced permanent Single Event Upsets once the injected charge exceeded the critical threshold, whereas the DICE SRAM automatically restored its original logic state after the transient disturbance. This improved performance is achieved through the redundant storage nodes and interlocked feedback mechanism of the DICE architecture. The results demonstrate that Radiation-Hardened-by-Design techniques significantly enhance memory reliability and establish the DICE SRAM as a robust solution for applications operating in radiation-prone and safety-critical environments.

CHAPTER 9

CONCLUSION

The primary objective of this project was to investigate the susceptibility of conventional Static Random Access Memory (SRAM) cells to radiation-induced Single Event Upsets (SEUs) and to evaluate the effectiveness of the Dual Interlocked Storage Cell (DICE) architecture as a Radiation-Hardened-by-Design (RHBD) solution. With the continuous scaling of CMOS technology, memory cells have become increasingly vulnerable to soft errors due to the reduction in supply voltage, node capacitance, and stored charge. Consequently, improving the reliability of SRAM has become an important research area, particularly for applications operating in radiation-prone environments.

The work began with the design and implementation of a conventional 6T SRAM cell using the Cadence Virtuoso Analog Design Environment in 180 nm CMOS technology. The transistor-level schematic was successfully developed, and the functionality of the memory cell was verified through transient simulations. The read, write, and hold operations were performed individually, and the obtained waveforms confirmed that the SRAM cell operated correctly under normal conditions. A precharge circuit was also designed and integrated with the SRAM cell to ensure proper initialization of the bit lines before every read operation. The successful implementation of the precharge circuit improved the reliability and accuracy of the read operation and provided the necessary conditions for subsequent radiation analysis.

Following the functional verification, the effects of radiation were investigated by injecting a double exponential transient current pulse into the sensitive storage nodes of the SRAM cell. The injected current pulse accurately represented the electrical effect produced by energetic particle strikes within semiconductor devices. By gradually increasing the magnitude of the injected current, the minimum threshold required to produce a Single Event Upset was identified. The corresponding critical charge values were calculated, providing a quantitative measure of the radiation tolerance of the conventional SRAM cell. The simulation results demonstrated that once the injected charge exceeded the critical threshold, the regenerative feedback of the cross-coupled inverters forced the SRAM cell into the opposite stable state, resulting in permanent data corruption. These observations confirmed the vulnerability of conventional SRAM cells to radiation-induced soft errors.

To improve memory reliability, a Dual Interlocked Storage Cell (DICE) SRAM architecture was subsequently designed and implemented using the same technology node. The DICE SRAM was subjected to the same functional verification procedures as the conventional SRAM. The read, write, and hold simulations confirmed that the DICE architecture maintained correct functionality under normal operating conditions while preserving compatibility with standard SRAM operations.

The radiation tolerance of the DICE SRAM was then evaluated using the same transient current injection methodology applied to the conventional SRAM. Unlike the conventional memory cell, the DICE architecture exhibited remarkable resistance to Single Event Upsets. Although the injected current pulse produced a temporary voltage disturbance at the affected storage node, the remaining interlocked storage nodes continued preserving the correct logic state. As the transient pulse disappeared, the regenerative feedback among the storage nodes automatically restored the disturbed node to its original value. No permanent bit flip was observed during the simulation, demonstrating the self-recovery capability of the DICE architecture.

The comparative analysis performed throughout this project clearly establishes the superiority of the DICE SRAM architecture in terms of radiation tolerance. While the conventional 6T SRAM successfully performs normal memory operations, it remains susceptible to radiation-induced data corruption once the critical charge threshold is exceeded. In contrast, the DICE SRAM significantly enhances reliability through redundant storage nodes and interlocked feedback paths, allowing the memory cell to withstand transient disturbances without external correction mechanisms. This self-correcting behavior makes the DICE architecture particularly suitable for applications requiring extremely high reliability.

The outcomes of this project demonstrate the importance of Radiation-Hardened-by-Design techniques in modern integrated circuit design. Rather than relying on specialized fabrication technologies, the DICE architecture achieves improved reliability through intelligent circuit design, making it a practical and efficient solution for mitigating Single Event Upsets. The implementation presented in this work provides a clear demonstration of how architectural modifications can substantially improve memory robustness while maintaining normal operating functionality.

In conclusion, this project successfully achieved all of its intended objectives. A conventional 6T SRAM cell was designed and verified, its susceptibility to radiation-induced Single Event Upsets was analyzed through transient current injection, the critical charge threshold responsible for data corruption was determined, and a radiation-hardened DICE SRAM architecture was successfully implemented and evaluated. The simulation results conclusively demonstrate that the DICE SRAM provides significantly greater resistance to Single Event Upsets through its redundant interlocked storage mechanism and automatic recovery capability. Therefore, the DICE architecture represents an effective and reliable solution for future memory systems intended for aerospace, satellite, defense, automotive, medical, and other safety-critical applications where dependable operation under radiation exposure is essential.

CHAPTER 10

FUTURE WORK

The work presented in this project demonstrates that the Dual Interlocked Storage Cell (DICE) architecture provides significantly improved resistance to Single Event Upsets compared to the conventional 6T SRAM cell. Although the objectives of this project were successfully achieved through schematic-level design and transient simulations, there remain several opportunities for extending this work to further improve the reliability, efficiency, and practical implementation of radiation-hardened memory systems.

One important extension of this work is the physical layout implementation of both the conventional 6T SRAM and the DICE SRAM architectures. In this project, the analysis was limited to schematic-level simulations. Designing the complete transistor layout using standard CMOS design rules would provide a more realistic representation of the circuit and enable investigation of parasitic capacitances, routing effects, and area utilization. Layout implementation would also allow evaluation of silicon area overhead introduced by the DICE architecture and provide insight into the trade-off between increased reliability and chip area.

Following layout design, post-layout simulation can be performed by extracting the parasitic resistance and capacitance associated with the fabricated layout. These parasitic elements influence the switching speed, propagation delay, power consumption, and stability of the memory cell. Performing post-layout transient simulations would provide more accurate performance estimates and help determine whether the radiation tolerance observed at the schematic level is maintained after considering real physical effects.

Another valuable extension involves Monte Carlo analysis. Semiconductor manufacturing introduces unavoidable variations in transistor dimensions, threshold voltages, oxide thickness, and doping concentrations. These process variations can significantly influence the stability and performance of SRAM cells. Monte Carlo simulations can be used to statistically evaluate the effects of these variations on read stability, write margin, hold stability, and critical charge. Such an analysis would provide a comprehensive understanding of the robustness of both conventional and radiation-hardened SRAM architectures under manufacturing uncertainties.

Future work may also include detailed investigation of process variation across different manufacturing corners. Parameters such as transistor threshold voltage, channel length, mobility, and supply voltage can be varied to evaluate circuit performance under best-case, worst-case, and typical operating conditions. Corner analysis would enable designers to identify operating conditions that are most susceptible to radiation-induced failures and optimize the circuit accordingly.

The radiation tolerance of the memory cell can also be investigated using more advanced semiconductor technology nodes. Although this project utilized a 180 nm CMOS process for simplicity and accurate circuit analysis, modern integrated circuits are fabricated using significantly smaller technologies such as 90 nm, 65 nm, 45 nm, 28 nm, and below. As transistor dimensions continue to shrink, the stored charge at internal nodes decreases, increasing the probability of radiation-induced soft errors. Repeating the proposed work using smaller technology nodes would provide valuable insight into the challenges associated with future semiconductor technologies.

Another promising direction is the implementation of the DICE architecture using FinFET technology. FinFET devices provide superior electrostatic control over the channel, reduced leakage current, improved switching characteristics, and better scalability compared to conventional planar CMOS transistors. Investigating radiation-hardened SRAM cells implemented using FinFET technology could further enhance reliability while reducing power consumption and improving performance in advanced integrated circuits.

Future research may also compare the DICE architecture with Error Correcting Codes (ECC). ECC techniques detect and correct memory errors by introducing redundant parity information into stored

data. A comparative analysis between architectural hardening through DICE and algorithmic error correction through ECC would provide a better understanding of their respective advantages, hardware overheads, power consumption, correction capabilities, and suitability for different applications. Furthermore, hybrid approaches combining DICE SRAM with ECC may provide even higher levels of fault tolerance for mission-critical systems.

Another interesting extension involves comparing the proposed DICE SRAM with Triple Modular Redundancy (TMR). TMR improves reliability by replicating the same circuit three times and using majority voting to determine the correct output. Although TMR offers excellent fault tolerance, it introduces significant area and power overhead. A detailed comparison between DICE SRAM and TMR in terms of radiation tolerance, silicon area, power consumption, propagation delay, implementation complexity, and reliability would provide valuable guidance for selecting the most appropriate radiation-hardening technique for different applications.

Further improvements can also be achieved by investigating additional Radiation-Hardened-by-Design (RHBD) techniques such as Quatro cells, hardened latches, capacitive hardening, guard rings, enclosed layout transistors, and node separation strategies. These techniques may be individually evaluated or combined with the DICE architecture to further improve immunity against multiple-node upsets and high-energy particle strikes.

Experimental validation of the proposed design represents another significant area for future work. Instead of relying solely on simulation-based current injection, fabricated SRAM chips could be exposed to controlled radiation environments using heavy-ion accelerators, neutron beams, or alpha-particle sources. Such experiments would validate the simulation results and provide a more comprehensive assessment of the effectiveness of the DICE architecture under real operating conditions.

The increasing demand for reliable electronic systems in aerospace, satellite communication, autonomous vehicles, defense electronics, nuclear instrumentation, industrial automation, and medical devices will continue to drive research in radiation-hardened memory design. The DICE architecture investigated in this project provides a strong foundation for future developments in this field. By extending the present work through physical implementation, advanced technology nodes, statistical analysis, and comparison with other fault-tolerant techniques, more efficient and highly reliable memory systems can be developed for next-generation integrated circuits.

In summary, this project establishes a foundation for future research on radiation-hardened SRAM design. The proposed extensions will enable a more comprehensive evaluation of the DICE architecture while addressing practical implementation challenges associated with modern semiconductor technologies. Continued research in this area will contribute to the development of robust memory systems capable of maintaining reliable operation in increasingly demanding and radiation-intensive environments.

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